



PROJECT TITLE -SEASONAL AGRICULTURE PERFORMANCE ANALYSIS

STUDENT NAME: THARIKA R

COLLEGE NAME: HINDUSTHAN COLLEGE OF TECHNOLOGY

AICTE STU ID : STU6a0e9e23df3221779342883



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

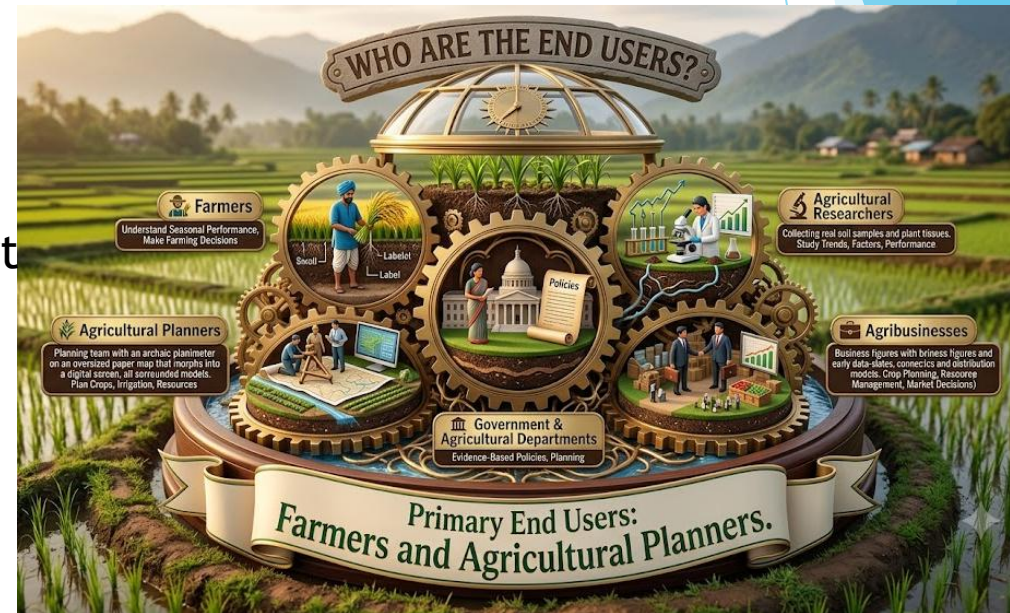


Seasonal Agriculture Performance Analysis is a data analytics project that studies how agricultural performance changes across **Kharif, Rabi, and Zaid seasons**. It analyzes crop yield, production, environmental conditions, irrigation, resource usage, disease/pest risk, revenue, cost, and profit. Using **Python, Pandas, NumPy, Matplotlib, Seaborn, and statistical techniques**, the project identifies seasonal patterns, relationships, and unusual observations. The findings provide **data-driven insights and recommendations** for better crop selection, resource management, irrigation, risk management, and seasonal agricultural planning.



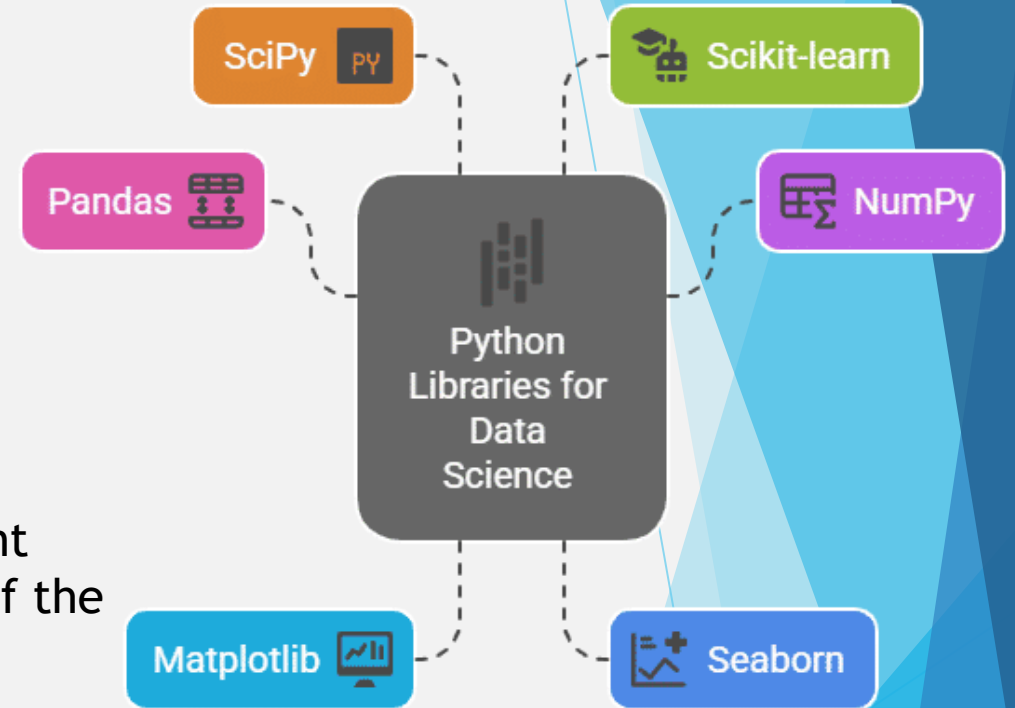
WHO ARE THE END USERS?

- ❑ **Farmers** - To understand seasonal crop performance and make better farming decisions.
- ❑ **Agricultural Planners** - To plan crops, irrigation, and resources based on seasonal patterns.
- ❑ **Government & Agricultural Departments** - To support evidence-based agricultural policies and planning.
- ❑ **Agricultural Researchers** - To study seasonal trends, environmental factors, and crop performance.
- ❑ **Agribusinesses** - To support crop planning, resource management, and market decisions.



Technology Used

- ❑ **Python** - Core programming and data analysis
- ❑ **Pandas** - Data cleaning, manipulation, and analysis
- ❑ **NumPy** - Numerical computations
- ❑ **Matplotlib** - Data visualization
- ❑ **Seaborn** - Statistical charts and heatmaps
- ❑ **SciPy** - Statistical testing and correlation analysis
- ❑ **Google Colab** - Development and execution environment
- ❑ **Jupyter Notebook** - Documentation and presentation of the analysis

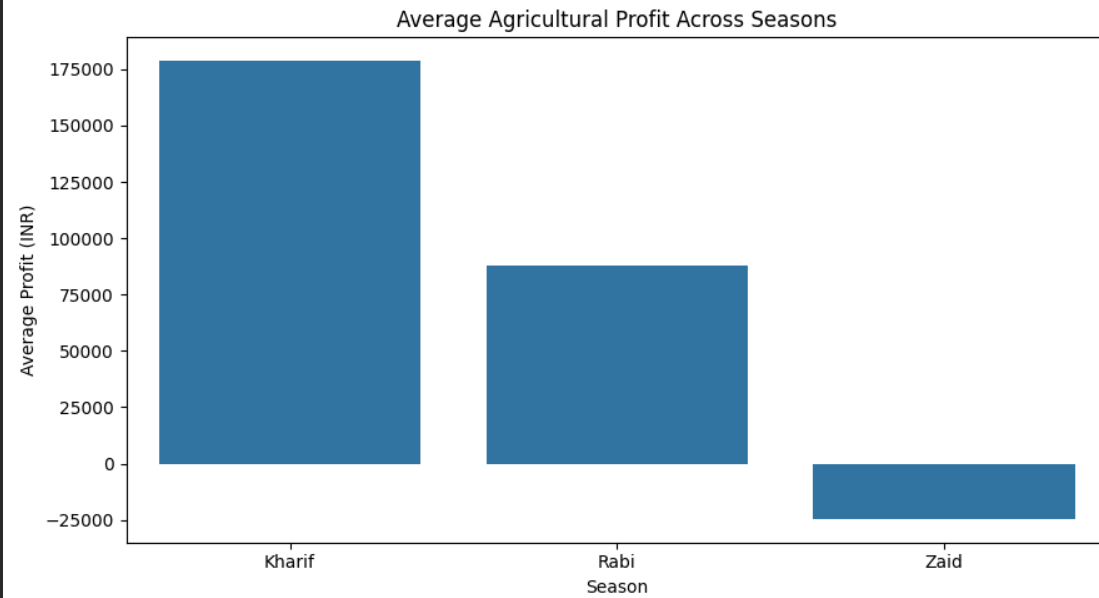


RESULTS

```
plt.figure(figsize=(9, 5))

sns.barplot(
    data=seasonal_performance.reset_index(),
    x='Season',
    y='Average_Profit'
)

plt.title("Average Agricultural Profit Across Seasons")
plt.xlabel("Season")
plt.ylabel("Average Profit (INR)")
plt.tight_layout()
plt.show()
```



RESULTS

```
[ ] ▶ plt.figure(figsize=(12, 7))

sns.heatmap(
    crop_season_yield,
    annot=True,
    fmt='.2f',
    cmap='YlGnBu'
)

plt.title("Crop Yield Performance Across Seasons")
plt.xlabel("Season")
plt.ylabel("Crop")
plt.tight_layout()
plt.show()
```

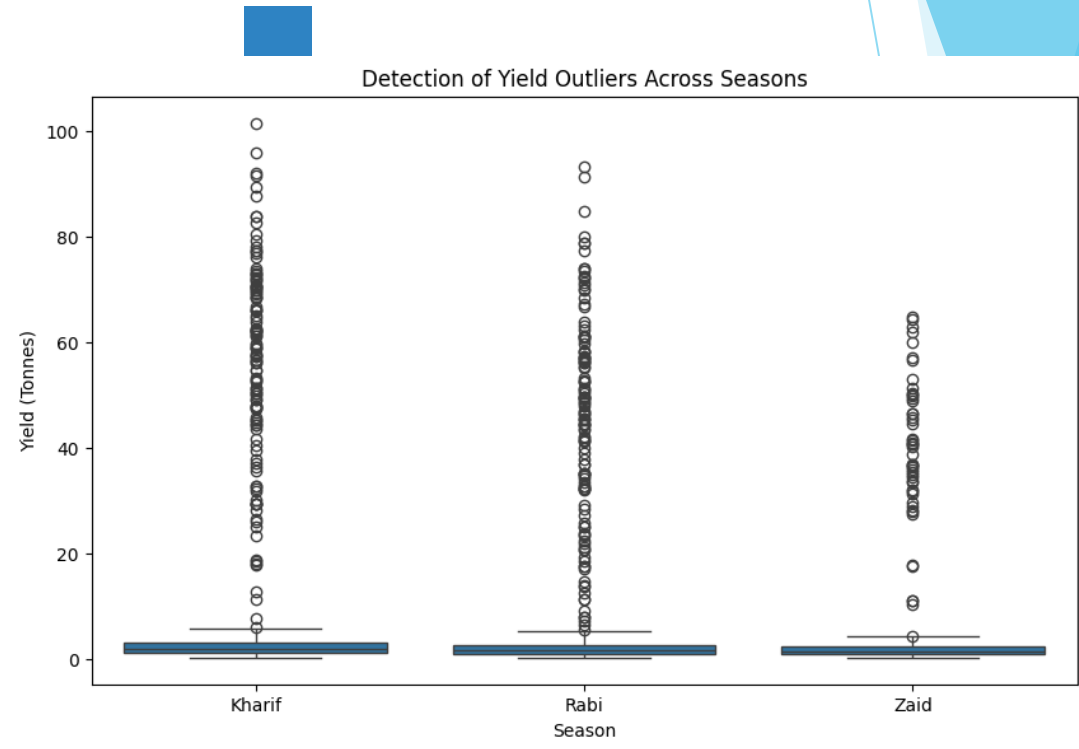


RESULTS

```
plt.figure(figsize=(10, 6))

sns.boxplot(
    data=df_clean,
    x='Season',
    y='Yield_Tonnes_Ha'
)

plt.title("Detection of Yield Outliers Across Seasons")
plt.xlabel("Season")
plt.ylabel("Yield (Tonnes)")
plt.show()
```



RESULTS

[Add screen shots of your code]

```
outlier_summary = yield_outliers[
    [
        'Season',
        'Crop',
        'Yield_Tonnes_Ha',
        'Production_Tonnes',
        'Rainfall_mm',
        'Water_Used_m3',
        'Disease_Pest_Risk_pct',
        'Profit_INR'
    ]
].sort_values('Yield_Tonnes_Ha', ascending=False)

outlier_summary.head(20)
```

...	Season	Crop	Yield_Tonnes_Ha	Production_Tonnes	Rainfall_mm	Water_Used_m3	Disease_Pest_Risk_pct	Profit_INR
1715	Kharif	Sugarcane	101.44	641.10	689.0	11781	60.1	1608291
3423	Kharif	Sugarcane	95.79	1424.40	745.9	23289	42.7	3330582
3525	Rabi	Sugarcane	93.13	648.18	772.5	10705	59.9	1471529
1464	Kharif	Sugarcane	91.98	703.65	579.6	12787	44.5	2216708
3523	Kharif	Sugarcane	91.46	75.00	752.4	1944	54.4	234320
2782	Rabi	Sugarcane	91.32	522.35	688.9	12199	50.6	1236801
644	Kharif	Sugarcane	89.34	668.26	797.2	11547	36.9	1984818
2847	Kharif	Sugarcane	87.76	672.24	703.0	12526	67.8	2041678
357	Rabi	Sugarcane	84.75	1249.21	589.1	23936	43.8	3378879
1178	Kharif	Sugarcane	83.79	534.58	941.4	11086	54.2	1200815
811	Kharif	Sugarcane	83.78	703.75	741.3	21051	54.8	2298743
3425	Kharif	Sugarcane	82.68	1154.21	879.1	23472	64.6	3740099
2019	Kharif	Sugarcane	80.36	581.00	679.6	13375	41.3	1567947

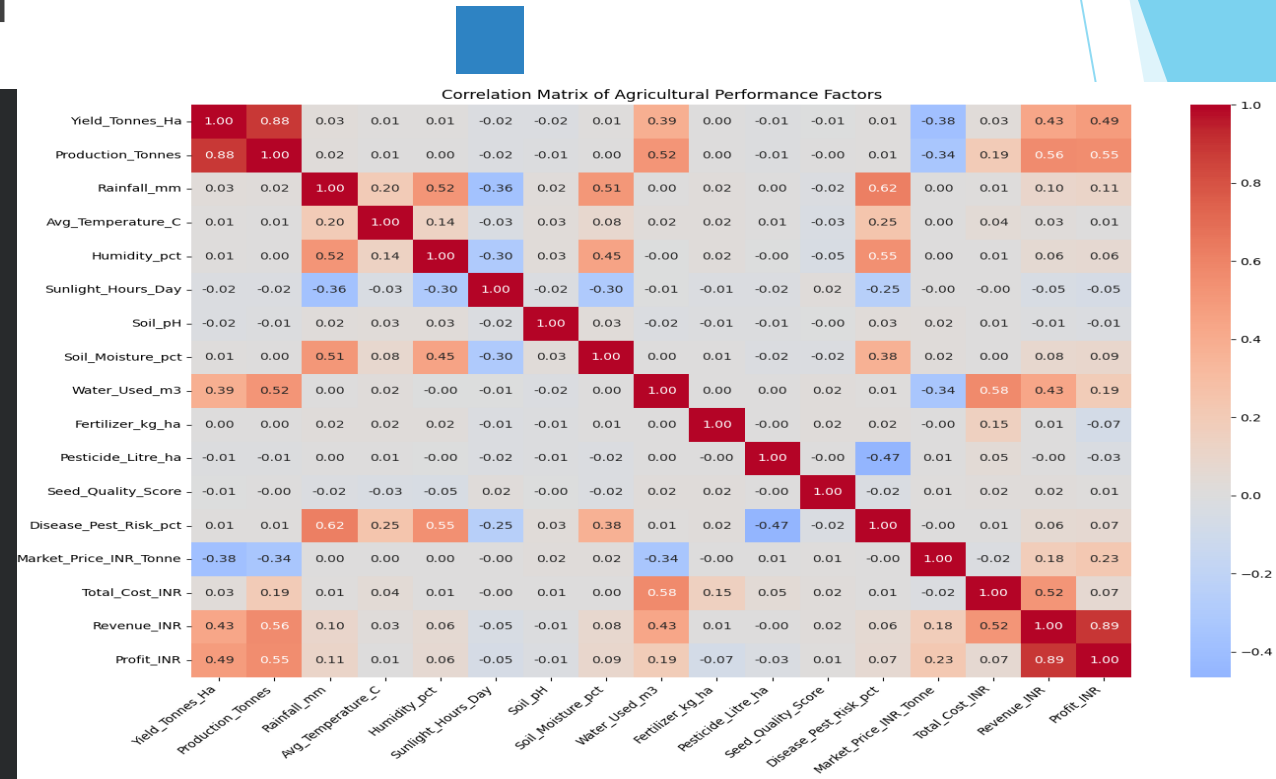
RESULTS

[Add screen shots of your code]

```
plt.figure(figsize=(14, 10))
```

```
sns.heatmap(  
    correlation_matrix,  
    annot=True,  
    fmt='.2f',  
    cmap='coolwarm',  
    center=0  
)
```

```
plt.title("Correlation Matrix of Agricultural Performance Factors")  
plt.xticks(rotation=45, ha='right')  
plt.yticks(rotation=0)  
plt.tight_layout()  
plt.show()
```



Future scope



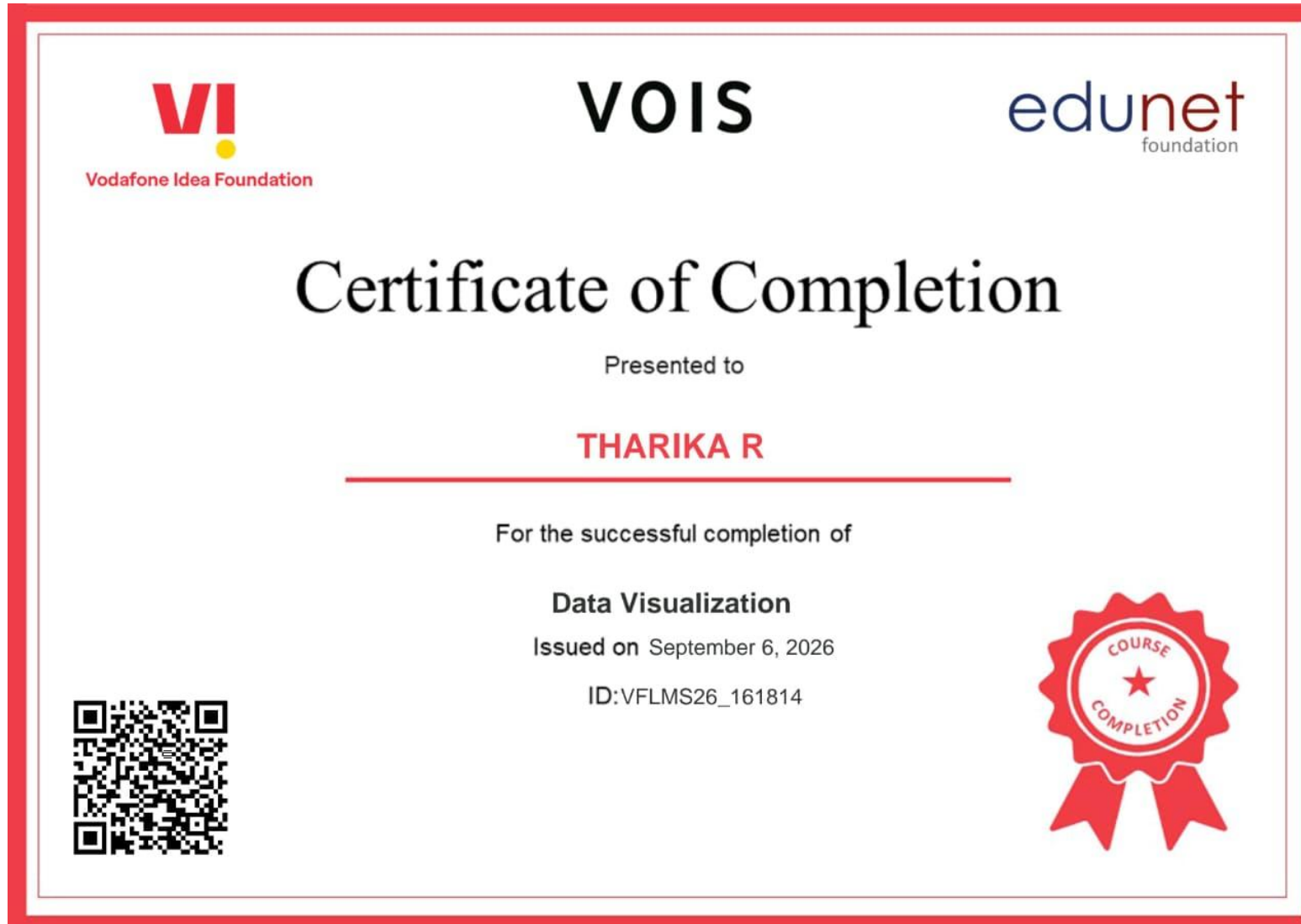
- ❖ **Real-Time Weather Integration** - Incorporate live weather data for better seasonal analysis.
- ❖ **Yield Prediction** - Use Machine Learning to predict crop yield for upcoming seasons.
- ❖ **Crop Recommendation** - Recommend suitable crops based on soil, climate, and seasonal conditions.
- ❖ **Smart Irrigation** - Develop intelligent irrigation recommendations to optimize water usage.
- ❖ **Disease & Pest Prediction** - Predict potential disease and pest risks using environmental data.
- ❖ **Interactive Dashboard** - Develop a user-friendly dashboard for farmers and agricultural planners.
- ❖ **Market Price Forecasting** - Predict future crop prices and profitability.
- ❖ **Satellite Data Integration** - Incorporate remote-sensing data for advanced crop and field monitoring.



GitHub Link

<https://github.com/tharikar87aids-png/SEASONAL-AGRICULTURE-PERFORMANCE-ANALYSIS->

VOIS Course completion certificate - Course name- Data Visualization



Thank you